Experiment Tracking (Weights & Biases)

All experiments for this assignment were tracked using Weights & Biases (WandB).

Separate runs were created for each notebook (Parts A–C), logging inference parameters, validation metrics (macro F1), and test set statistics.

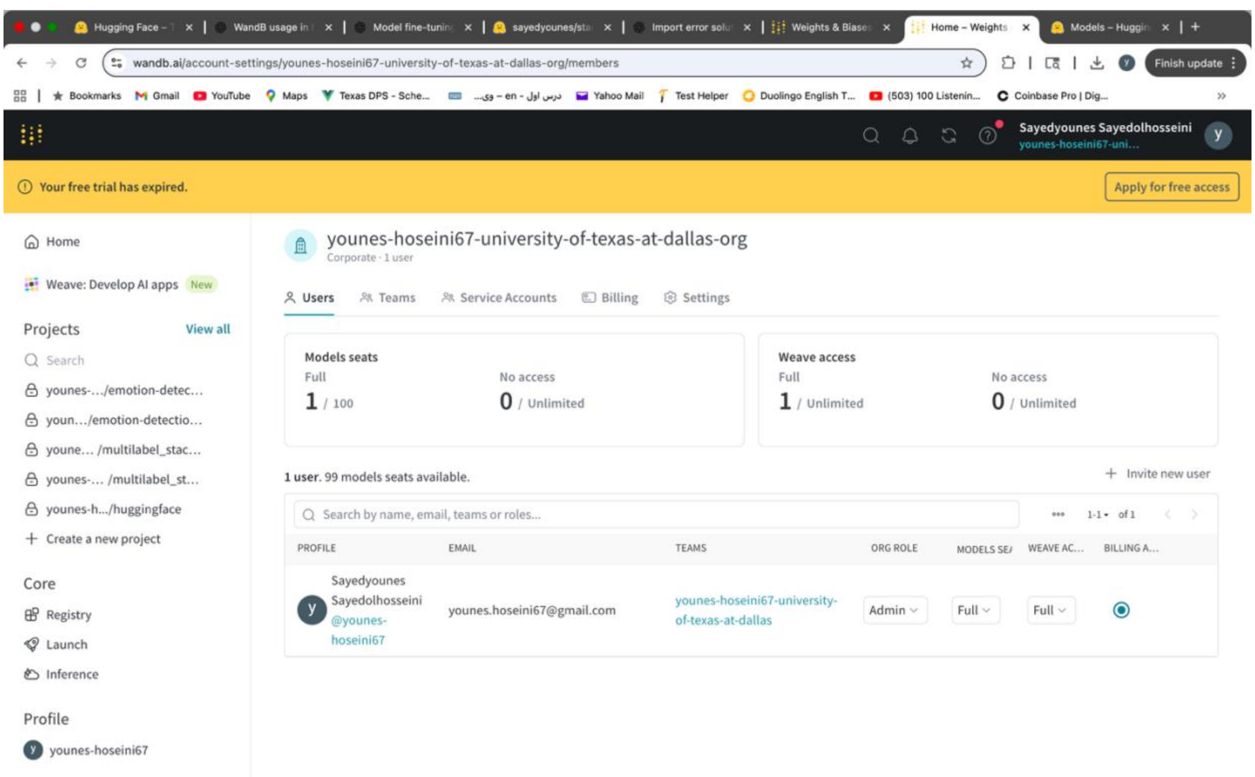
Public WandB Project:

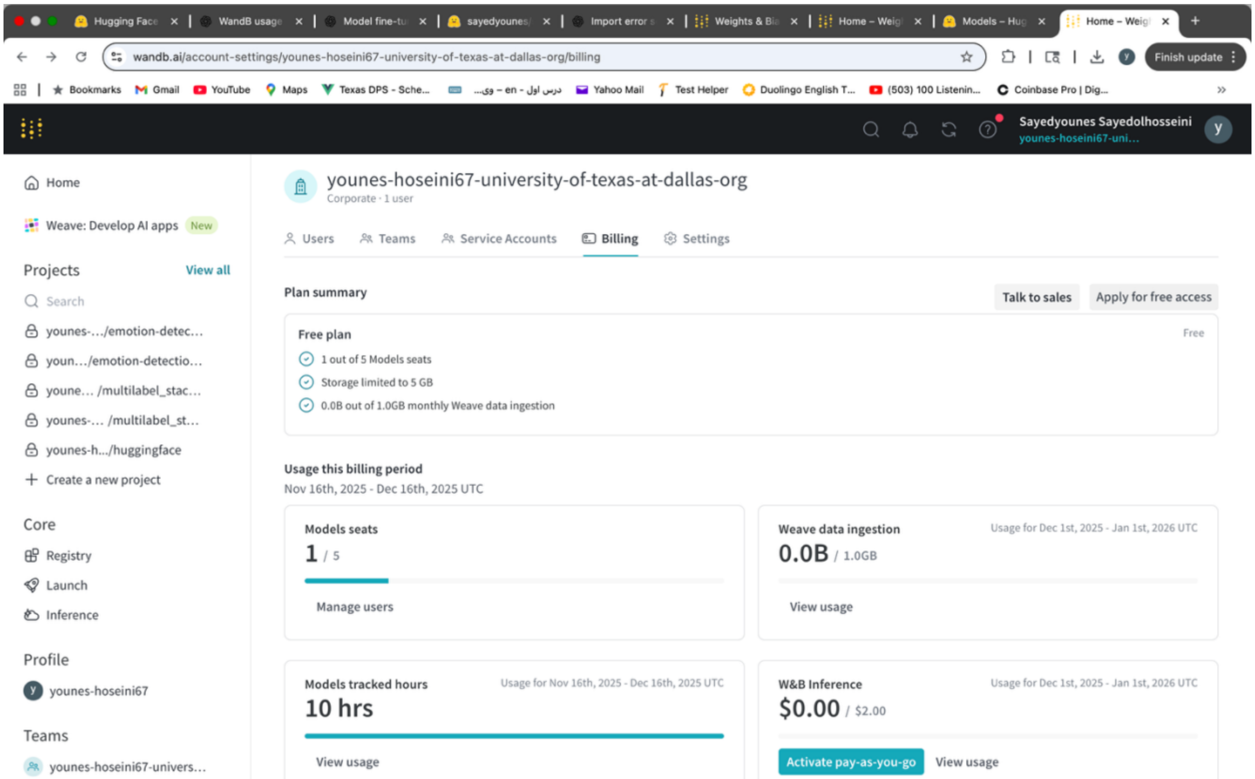
https://wandb.ai/account-settings/younes-hoseini67-university-of-texas-at-dallas-org/members

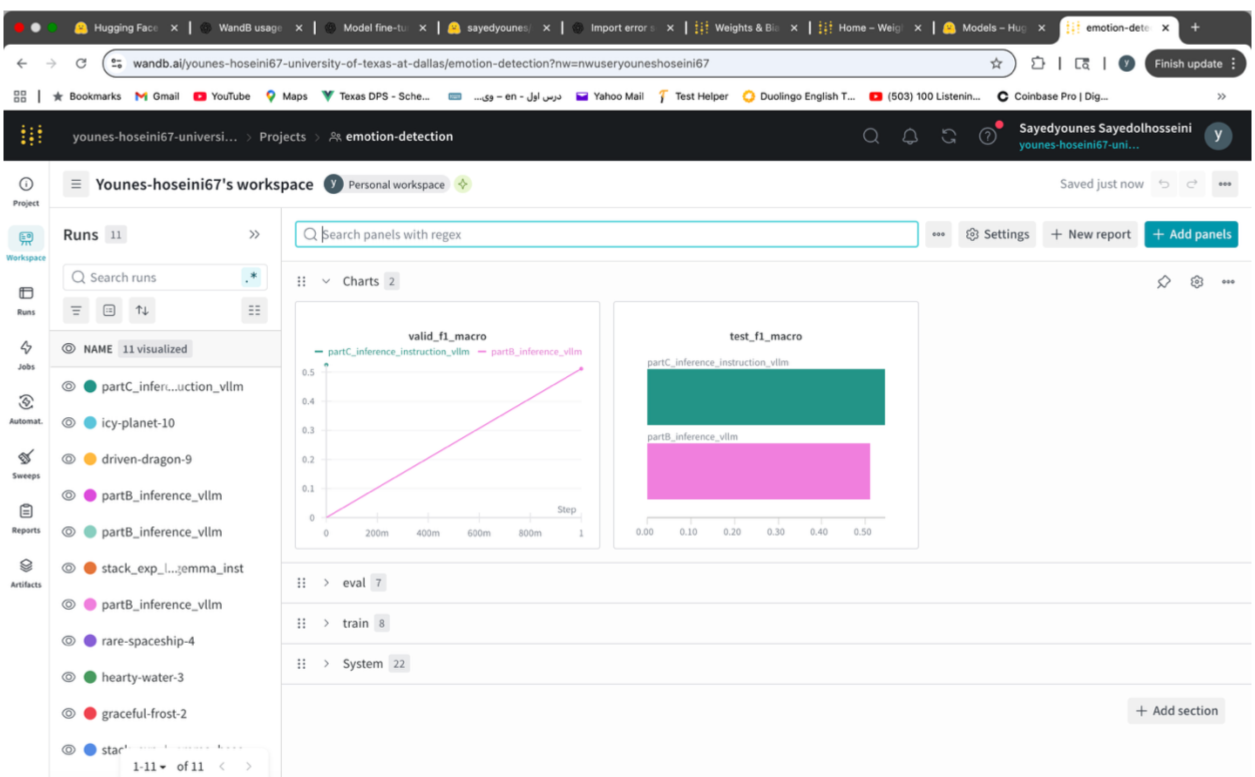
link for one of project:

https://wandb.ai/younes-hoseini67-university-of-texas-at-dallas/emotion-detection?nw=nwuseryouneshoseini67

screenshot of main page:







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Summary and README:

Multilabel Emotion Classification with Gemma-2-2B and vLLM

Overview

This project demonstrates inference and evaluation of a decoder-only large language model for a multilabel text classification task. The goal is to predict one or more emotion labels for each input text using a base instruction-style language model and high-performance inference.

The workflow follows the required pipeline:

dataset loading, model inference with guided decoding, validation using multilabel metrics, and test-set prediction for Kaggle submission.

Dataset

We use the StackExchange Emotion dataset, where each text sample may belong to multiple emotion categories. The task is formulated as a multilabel classification problem with the following 11 emotion classes:

- anger
- anticipation
- disgust
- fear
- joy
- love
- optimism
- pessimism
- sadness
- surprise
- trust

Model

The base **Gemma-2-2B** decoder-only language model (`google/gemma-2-2b`) is used without

fine-tuning. The model is prompted using an instruction-based format to perform emotion

classification.

Part A – Classification Head

- Fine-tuned decoder model with classification head
- Binary outputs for each emotion
- Notebook: `PartA_Classification_Head.ipynb`

🔷 Part B – Language Modeling Head

- Reformulated classification as text generation
- Labels generated as text (e.g., "complies" / "violates")
- Used constrained decoding

Notebooks:

- `PartB_Training_Language_Head.ipynb`
- `PartB_Inference_vLLM.ipynb`

🔷 Part C – Instruction-Tuned Model

- Instruction-based prompting with decoder model
- Generates structured outputs (emotion labels)

Notebooks:

- `PartC_Training_Instruction_Model.ipynb`
- `PartC_Inference_Instruction_vLLM.ipynb`

Inference with vLLM

Inference is performed using **vLLM**, enabling efficient batch generation and scalable

evaluation. Prompts are constructed in an instruction format, and predictions are generated

in batches to improve throughput.

Guided Decoding

To ensure valid and structured outputs, **JSON-guided (constrained) decoding** is applied. The model is instructed to return emotion labels strictly as a JSON array. This reduces invalid outputs and simplifies downstream parsing for multilabel evaluation.

Evaluation

Model performance is evaluated on a validation set using **macro F1 score**, which is appropriate for multilabel classification with class imbalance. Confusion matrices are used for qualitative analysis of per-class prediction behavior.

Validation and test results are logged using **Weights & Biases (WandB)** for experiment tracking and reproducibility.

Test Set and Kaggle Submission

The trained inference pipeline is applied to the test dataset, and predictions are converted into the required Kaggle submission format with one binary column per emotion label. The final submission is evaluated on the Kaggle leaderboard.

Summary

This project demonstrates:

- Multilabel emotion classification using a decoder-only LLM
- High-performance inference with vLLM
- Structured output enforcement via guided decoding
- Proper multilabel evaluation using macro F1
- End-to-end test inference and Kaggle submission